BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 208 · Synoptic programming challenge

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Give the Big O of a full scan of n values.

Q2. Which item does a queue remove first?

Q3. What does return count do?

Q4. Starting total=0, sum even values from [3,4,6].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

decomposition — Break a problem into smaller tasks.

threshold — Value used as a cutoff.

counter — Variable that records how often an event occurs.

boundary test — Input at or next to a permitted limit.

erroneous input — An input violating the program's constraints.

New learning

Inputs and output — Inputs are an array and a threshold; the result is the number of qualifying items.

Check each element — The counter increases only for values greater than or equal to the threshold.

Test a boundary — A value equal to the threshold must count. For 5,8,3,8 at threshold 8, result is 2.

Worked example

Count values at least 7 in [4,7,9].

1. Predict — Skip 4 because it is below 7.

2. Explain — Count 7 and 9: the result is 2.

We Do · Practise with guidance

Count values at least 6 in [6,2,6,8].

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Threshold = 6; count starts at 0.
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Predict

The first 6 counts and 2 does not.

Explain

The second 6 and 8 count; final result is 3.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Find CountAtLeast([2,5,7],5).

Question 2

Change the method so it counts values strictly greater than the threshold.

Question 3

Give normal, boundary and erroneous test cases if the threshold must be non-negative.

Exit Ticket · Answer before checking

The test score>=t excludes values equal to t.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. O(n).

Q2. The oldest enqueued item.

Q3. Returns count to the caller.

Q4. 10 (4+6).

You Do pages · Independent questions

Q1. 2 because 5 and 7 qualify.

Q2. Change score>=t to score>t.

Q3. Normal threshold 5 with [3,5,7] gives 2; boundary threshold 0 with [0] gives 1; erroneous threshold −1 must be rejected using an input guard.

Exit Ticket · Model answer

False. >= includes equality.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: